

Docker:

Docker is a platform that provides virtual containers on which an application can be deployed independent of the underlying OS of the server. Further the container can be created from a replica called docker image which contains all the dependencies and can run on any OS that has docker engine, with similar results.

VIRTUALIZATION:

Virtualization is the process of sharing hardware resources across several virtually isolated and mutually independent systems. It is achieved by using a hypervisor which acts as a bridge between the Operating System of each of the virtual machines and the underlying hardware.

Applications in virtual environments run on a host operating system on top of the hypervisor.

BASIC DOCKER COMMANDS

? Display docker images available in our machine

```
$ docker images
```

? Download docker image.

```
$ docker pull <image-name / image-id>
```

? Run docker image.

```
$ docker run <image-name / image-id>
```

? Delete docker image.

```
$ docker rmi <image-name / image-id>
```

? Display all running docker containers.

```
$ docker ps
```

? Display all running and stopped containers.

```
$ docker ps -a
```

? Delete docker container.

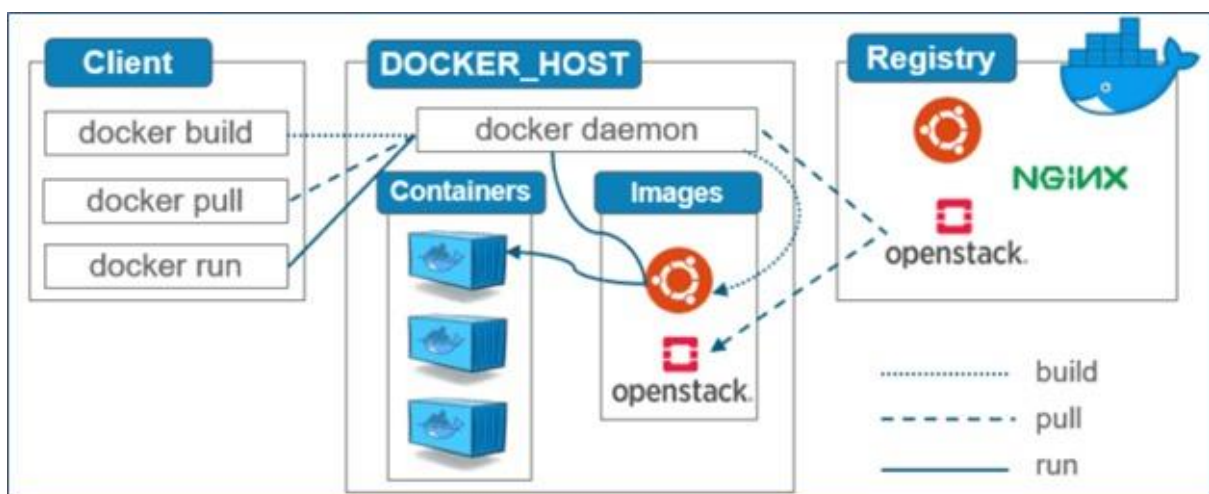
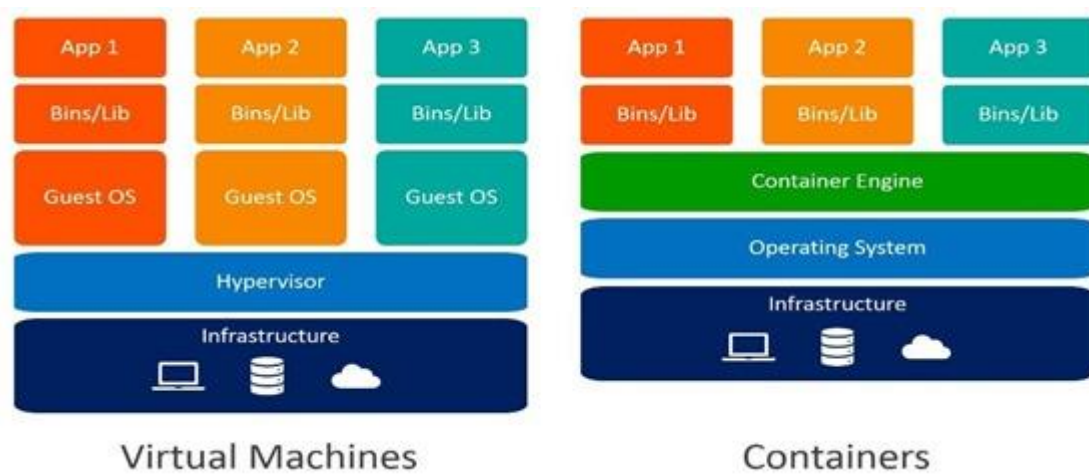
```
$ docker rm <container-id>
```

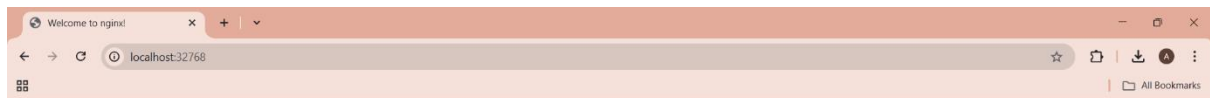
❓ Delete docker image forcefully.

```
$ docker rmi -f <image-id>
```

❓ Stop Docker container.

```
$ docker stop <container-id>
```





Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Images [Give feedback](#)

View and manage your local and Docker Hub images. [Learn more](#)

Local **Hub repositories**

206.45 MB / 81.67 MB in use 1 images Last refresh: 14 minutes ago

Search

	Name	Tag	Image ID	Created	Size	Actions
☐	nginx	latest	706959c57d76	1 month ago	278.62 MB	Play Info Delete

Terminal

```
Windows PowerShell
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PS C:\Users\ASHW\NT> docker run -itd -P sha256:706959c57d76560ed77ea37b7e0a203380b5b12c153c4f820449b9d43d292f53
55801a39d0a013acbec3efca6f6818543c7f5a8008f9234f312c7e9b8d0620532
PS C:\Users\ASHW\NT> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                    NAMES
55801a39d0a0   706959c57d76                        "/docker-entrypoint..." 17 seconds ago Up 16 seconds   0.0.0.0:32768->80/tcp   unruffled_ferret
```

Engine running | RAM: 0.95 GB CPU: 0.00% Disk: 1.49 GB used (limit 1006.85 GB) Terminal v4.38.0